

Simple Model: Stability of a Microscopic Black Hole Reactor: Hawking Evaporation versus Bondi Accretion

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Abstract

We analyse the mass dynamics of a microscopic black hole (MBH) placed in a dense medium, treating Hawking evaporation and Bondi accretion as competing processes governed by the equation $\dot{M} = AM^2 - B/M^2$, where the coefficients A and B are determined by the surrounding medium and fundamental constants respectively. We show analytically that this equation admits a unique positive fixed point $M_{\text{crit}} = (B/A)^{1/4}$ which is an *unstable* equilibrium (a repeller): the derivative $d\dot{M}/dM|_{M_{\text{crit}}} > 0$ for all physically admissible media. For a reference MBH of mass $M_0 = 10^7 \text{ kg}$ immersed in water, $M_{\text{crit}} \approx 3.12 \times 10^{10} \text{ kg} \approx 3120 M_0$: the reactor lies deep in the evaporation regime and expires in $\tau \approx 23.6 \text{ h}$. Numerical integration of the full ODE confirms that every sub-critical trajectory terminates in finite time, while every super-critical trajectory diverges — but on a timescale $\tau_{\text{grow}} \sim 10^{10} \text{ yr}$, rendering the accretion regime equally impractical. A parameter-space analysis shows that no physically realistic medium lowers M_{crit} to M_0 , establishing the fundamental impossibility of long-term MBH stabilisation via Bondi accretion.

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1 Introduction

The idea of harnessing a black hole as an energy source has captured both scientific and popular imagination since Hawking’s discovery that black holes emit thermal radiation [1]. A microscopic black hole (MBH) of mass $M \sim 10^7$ kg radiates at an extraordinary power of $\sim 3.5 \times 10^{18}$ W—roughly 3.5×10^9 times the output of a modern nuclear reactor—while converting its entire rest mass into radiation with 100% efficiency. The engineering fantasy of a “black hole reactor” therefore has genuine physical motivation.

The fatal obstacle is twofold. First, Hawking evaporation is a runaway process: as the MBH loses mass, its temperature $T \propto 1/M$ rises, accelerating emission until the object evaporates explosively. An $M_0 = 10^7$ kg MBH survives for only $\tau \approx 23.6$ h. Second, the natural stabilisation candidate—Bondi accretion of surrounding matter—turns out to be insufficient, not merely in magnitude but in principle.

In this paper we derive and analyse the complete mass equation:

$$\dot{M} = AM^2 - \frac{B}{M^2}, \quad (1)$$

where A encodes Bondi accretion from the medium and B encodes Hawking evaporation through fundamental constants. We prove that the unique fixed point of (1) is an unstable repeller, provide numerical solutions for the full trajectory, and map the stability boundary across all physically meaningful media.

The problem was originally formulated as an olympiad problem for the Kyrgyz Republic national physics competition serving as an IPhO-2026 selection round (Day 2, Problem 3, April 2026). The present paper extends the olympiad treatment to a complete dynamical systems analysis, adding the phase portrait, full numerical integration, and the medium-dependence study that appear as Figures 1–3.

2 Physical Model

2.1 Hawking Evaporation

The Schwarzschild radius of a black hole of mass M follows from the escape-velocity condition $v_{\text{esc}} = c$:

$$R_s = \frac{2GM}{c^2}. \quad (2)$$

For $M_0 = 10^7$ kg, $R_s \approx 1.48 \times 10^{-20}$ m, five orders of magnitude smaller than a proton.

The Hawking temperature is obtained by treating a photon born near the horizon as localised in a region $\Delta x \approx 2R_s$ and applying the Heisenberg uncertainty principle $\Delta x \Delta p \sim \hbar/2$, giving a characteristic energy $\varepsilon \sim \Delta p \cdot c$. Setting $\varepsilon = k_B T$ with the exact

coefficient $(8\pi)^{-1}$ from quantum field theory in curved spacetime [1]:

$$T(M) = \frac{\hbar c^3}{8\pi G M k_B}. \quad (3)$$

The radiated power follows from the Stefan–Boltzmann law $P = \sigma A T^4$ with horizon area $A = 4\pi R_s^2 = 16\pi G^2 M^2 / c^4$:

$$P(M) = \frac{K}{M^2}, \quad K = \frac{\sigma \hbar^4 c^8}{256\pi^3 G^2 k_B^4} \approx 3.53 \times 10^{32} \text{ W kg}^2. \quad (4)$$

The mass loss due to radiation is $P = -c^2 \dot{M}$, so:

$$\dot{M}|_{\text{evap}} = -\frac{B}{M^2}, \quad B = \frac{K}{c^2} \approx 3.92 \times 10^{15} \text{ kg}^3 \text{ s}^{-1}. \quad (5)$$

Integrating with $M(0) = M_0$ and $M(\tau) = 0$ gives the evaporation lifetime:

$$\tau = \frac{M_0^3 c^2}{3K} \approx 8.50 \times 10^4 \text{ s} \approx 23.6 \text{ h}. \quad (6)$$

2.2 Bondi Accretion

A molecule of the surrounding medium (density ρ , sound speed v_s) is gravitationally captured if its kinetic energy is less than the gravitational potential energy at the Bondi radius [2]:

$$R_B = \frac{GM}{v_s^2}. \quad (7)$$

Treating all matter passing through the cross-section πR_B^2 as accreted and moving at speed v_s :

$$\dot{M}|_{\text{acc}} = \rho v_s \cdot \pi R_B^2 = \frac{\pi \rho G^2}{v_s^3} M^2 = AM^2, \quad (8)$$

where $A = \pi \rho G^2 / v_s^3$. For water ($\rho = 10^3 \text{ kg m}^{-3}$, $v_s = 1500 \text{ m s}^{-1}$): $A \approx 4.14 \times 10^{-27} \text{ kg}^{-1} \text{ s}^{-1}$.

2.3 Combined Dynamics

Combining (5) and (8):

$$\boxed{\dot{M} = AM^2 - \frac{B}{M^2}}. \quad (9)$$

The two terms scale oppositely with M : accretion grows as M^2 , evaporation decays as M^{-2} . At $M_0 = 10^7 \text{ kg}$ the evaporation rate is $\approx 39 \text{ kg s}^{-1}$ while the accretion rate is $\approx 4.1 \times 10^{-13} \text{ kg s}^{-1}$ —a ratio of $\sim 10^{14}$. Accretion is entirely negligible for the sub-critical reactor.

3 Stability Analysis

3.1 Fixed Point

Setting $\dot{M} = 0$ in (9):

$$AM^2 = \frac{B}{M^2} \implies M_{\text{crit}} = \left(\frac{B}{A}\right)^{1/4} = \left(\frac{K v_s^3}{\pi \rho G^2 c^2}\right)^{1/4}. \quad (10)$$

For water: $M_{\text{crit}} \approx 3.12 \times 10^{10} \text{ kg} \approx 3120 M_0$.

3.2 Instability Proof

The stability of M_{crit} is determined by the sign of $d\dot{M}/dM$ evaluated at the fixed point. Differentiating (9):

$$\frac{d\dot{M}}{dM} = 2AM + \frac{2B}{M^3}. \quad (11)$$

Both terms are strictly positive for all $M > 0$, so:

$$\left.\frac{d\dot{M}}{dM}\right|_{M_{\text{crit}}} = 2AM_{\text{crit}} + \frac{2B}{M_{\text{crit}}^3} > 0. \quad (12)$$

The fixed point M_{crit} is an unstable equilibrium (repeller) for every positive A , B and every positive mass.

The physical consequence is immediate:

- If $M < M_{\text{crit}}$: evaporation dominates, $\dot{M} < 0$, the MBH loses mass and evaporates completely in finite time.
- If $M > M_{\text{crit}}$: accretion dominates, $\dot{M} > 0$, the MBH grows without bound.

No perturbation from M_{crit} returns to it; the equilibrium repels all nearby trajectories.

3.3 Radiation Pressure Barrier

An independent confirmation of instability arises from the Eddington-like argument. The ratio of radiation pressure force to gravitational force on a proton at distance r is [3]:

$$\frac{F_{\text{rad}}}{F_{\text{grav}}} = \frac{\sigma_T K}{4\pi c G m_p M^3}, \quad (13)$$

which is independent of r . Setting this ratio to unity recovers M_{crit} up to a numerical factor of order unity, confirming that the dynamical M_{crit} and the Eddington-limit M_{crit} coincide. At M_0 , the ratio is $\approx 5.6 \times 10^{10}$ —radiation pressure overwhelms gravity by ten orders of magnitude.

4 Numerical Results

Equation (9) was integrated using a Runge–Kutta 4(5) adaptive solver with relative tolerance 10^{-9} . All computations were performed in Python using `scipy.integrate.solve_ivp`.

4.1 Phase Portrait

Figure 1 shows $\dot{M}$ versus M on a logarithmic mass axis. The left panel (absolute units) reveals that for all sub-critical masses $\dot{M}$ is large and negative (up to -39 kg s^{-1} at M_0), while the accretion contribution is negligible on this scale. The right panel normalises $\dot{M}$ by $AM_{\text{crit}}^2 \approx 4 \times 10^{-6} \text{ kg s}^{-1}$ and zooms near M_{crit} to reveal the repeller structure: the curve crosses zero with positive slope, with the two shaded regions corresponding to the two dynamical regimes.

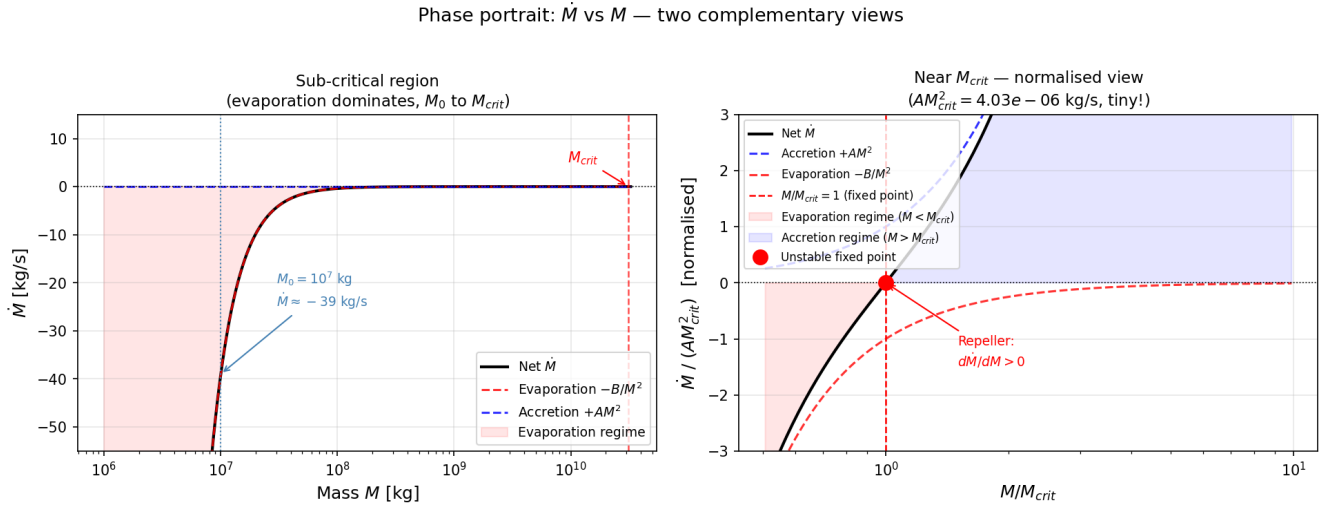


Figure 1: Phase portrait $\dot{M}(M)$. **(a)** Absolute units, sub-critical region. Evaporation (red dashed) dominates by many orders of magnitude over accretion (blue dashed) for $M \lesssim M_{\text{crit}}$. **(b)** Normalised units near M_{crit} . The net $\dot{M}$ (solid black) crosses zero with positive slope, confirming M_{crit} is an unstable fixed point (red circle). Red and blue shading indicate evaporation and accretion regimes respectively.

4.2 Mass Trajectories

Figure 2 shows numerically integrated trajectories for sub-critical (left) and super-critical (right) initial masses.

A key observation from Figure 2(b) is that the super-critical regime, while formally stable against evaporation, operates on geological timescales. The asymmetry between the two sides of the repeller is thus extreme: below M_{crit} the MBH evaporates in hours, above M_{crit} it doubles in $\sim 10^8$ years.

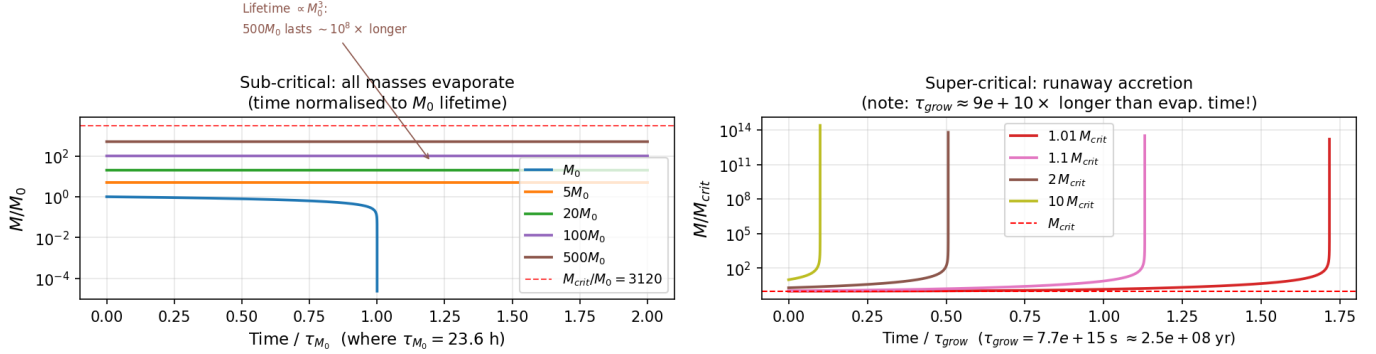


Figure 2: Mass trajectories $M(t)$. **(a)** All sub-critical initial masses evaporate completely. Time is normalised to $\tau_{M_0} = 23.6$ h; mass scales as $\tau \propto M_0^3$ (Eq. 6). **(b)** Super-critical masses grow, but the characteristic timescale $\tau_{grow} = 1/(AM_{crit}) \approx 2.5 \times 10^8$ yr is $\sim 10^{10}$ times longer than the evaporation lifetime, making the accretion-dominated regime equally impractical.

4.3 Medium Dependence

From (10), $M_{crit} \propto (v_s^3/\rho)^{1/4}$. Figure 3 maps $M_{crit}(\rho, v_s)$ across the full parameter space of physical media. The white contour marks $M_{crit} = M_0$: *only media above this contour could in principle stabilise a reactor of mass M_0* . Water (black star) lies far below this contour by three orders of magnitude in M_{crit} . No known or physically conceivable medium places an $M_0 = 10^7$ kg reactor in the stable region.

5 Discussion

5.1 Why Both Sides of the Fixed Point Fail

The instability of M_{crit} is not merely a statement that one particular initial mass leads to evaporation. It is a topological statement: the fixed point is a repeller, so the set of initial conditions that leads to a bounded, non-diverging trajectory has measure zero (exactly $\{M = M_{crit}\}$). Any real device subject to quantum fluctuations, thermal noise, or manufacturing imprecision will not remain at M_{crit} .

Moreover, even the formally “winning” super-critical side is physically unacceptable: a black hole growing at the Bondi rate from M_{crit} doubles its mass in $\tau_{grow} \sim 1/(AM_{crit}) \approx 2.5 \times 10^8$ yr. The instability is therefore double-sided: catastrophic on the sub-critical side (hours), irrelevant on the super-critical side (geological epochs).

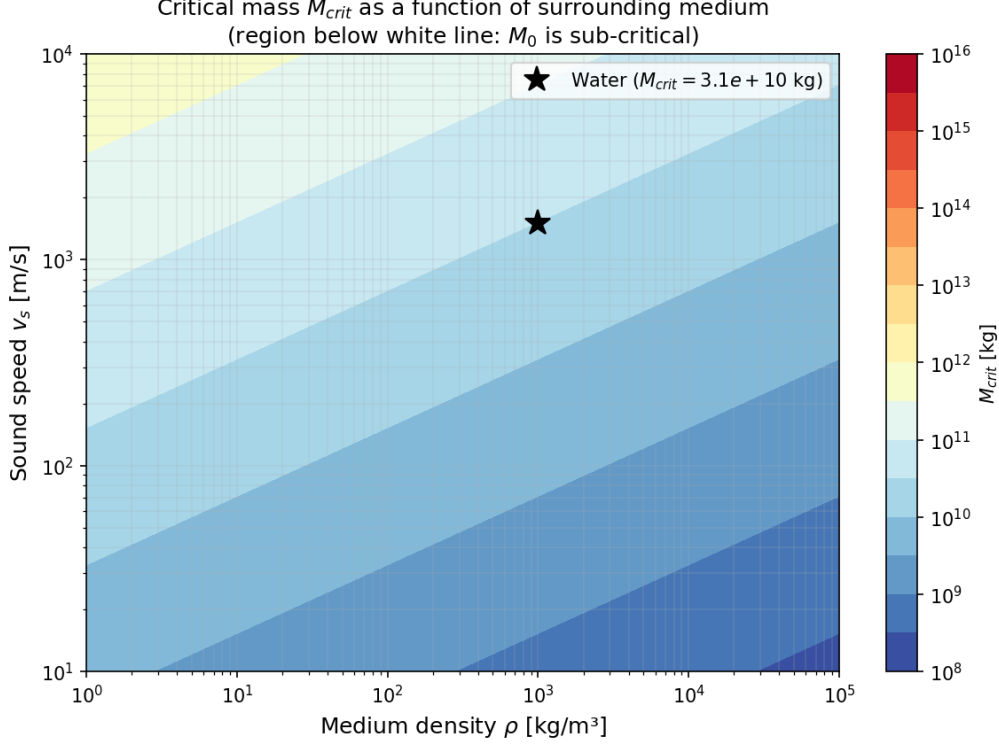


Figure 3: Critical mass $M_{crit}(\rho, v_s)$ as a function of medium density and sound speed. The white contour marks $M_{crit} = M_0 = 10^7$ kg. Only media above this line could stabilise the reference reactor. Water (black star) lies far below, with $M_{crit} \approx 3120 M_0$.

5.2 Connection to the Eddington Limit

The critical mass (10) derived from the dynamical balance $AM^2 = B/M^2$ coincides (up to a factor of order unity) with the mass at which radiation pressure equals gravitational attraction on an accreting proton. This is the Eddington limit for a black hole, a well-known concept in astrophysics [3]. The present analysis shows that the Eddington limit emerges naturally as the unstable fixed point of the evaporation-accretion ODE, providing a novel pedagogical derivation of this fundamental astrophysical concept.

5.3 Limitations

The model makes several simplifying assumptions. (i) The Bondi formula assumes a uniform, pressureless medium at rest; real fluid dynamics near a compact object is more complex [2]. (ii) The Hawking temperature formula (3) is derived in the semiclassical limit; quantum gravity corrections near the Planck mass ($\sim 10^{-8}$ kg) may alter the final evaporation stage. (iii) We neglect the momentum imparted to the MBH by anisotropic emission, which could affect accretion geometry. None of these corrections change the sign of $d\dot{M}/dM$, so the qualitative conclusion (instability) is robust.

6 Conclusion

We have shown that the mass dynamics of a microscopic black hole immersed in a dense medium is governed by $\dot{M} = AM^2 - B/M^2$, which admits a unique positive fixed point $M_{\text{crit}} = (B/A)^{1/4}$ that is *necessarily unstable* for all physically admissible parameters. The proof is algebraic: $d\dot{M}/dM = 2AM + 2B/M^3 > 0$ for all $M > 0$.

For a reference reactor mass $M_0 = 10^7$ kg in water, $M_{\text{crit}} \approx 3120 M_0$ and the evaporation lifetime is ≈ 23.6 h. Numerical integration confirms that all sub-critical trajectories terminate in finite time, while the super-critical growth timescale of $\sim 10^8$ yr renders the accretion-dominated regime equally impractical. A systematic parameter sweep shows that no known physical medium lowers M_{crit} to M_0 .

The long-term stabilisation of a microscopic black hole reactor via Bondi accretion is fundamentally impossible, not merely an engineering challenge.

7 Numerical Values of Key Parameters

Table 1: Key physical quantities for the reference reactor ($M_0 = 10^7$ kg, water medium).

Quantity	Symbol	Value	Unit
Schwarzschild radius	$R_s(M_0)$	1.48×10^{-20}	m
Hawking temperature	$T(M_0)$	1.23×10^{16}	K
Radiation power	$P(M_0) = K/M_0^2$	3.53×10^{18}	W
Evaporation lifetime	τ	23.6	h
Total energy released	$E = M_0 c^2$	9.00×10^{23}	J
Bondi radius	$R_B(M_0)$	2.96×10^{-10}	m
Accretion rate	$\dot{M}_{\text{acc}}(M_0)$	4.14×10^{-13}	kg s ⁻¹
Evaporation rate	$\dot{M}_{\text{evap}}(M_0)$	39.2	kg s ⁻¹
Evap./acc. ratio	—	$\approx 10^{14}$	—
Critical mass	M_{crit}	3.12×10^{10}	kg
M_{crit}/M_0	—	≈ 3120	—
Growth timescale	$\tau_{\text{grow}} = 1/(AM_{\text{crit}})$	2.5×10^8	yr

8 Derivation of the Evaporation Constant K

Substituting $A_{\text{horizon}} = 16\pi G^2 M^2 / c^4$ and $T^4 = \hbar^4 c^{12} / [(8\pi)^4 G^4 M^4 k_B^4]$ into $P = \sigma A_{\text{horizon}} T^4$:

$$P = \sigma \cdot \frac{16\pi G^2 M^2}{c^4} \cdot \frac{\hbar^4 c^{12}}{(8\pi)^4 G^4 M^4 k_B^4} = \frac{\sigma \hbar^4 c^8}{256\pi^3 G^2 k_B^4} \cdot \frac{1}{M^2} = \frac{K}{M^2}, \quad (14)$$

giving $K \approx 3.53 \times 10^{32}$ W kg².

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